



The "Google" distance

[Bookmark this page](#)

Discussions

All posts sorted by recent activity

Problem: Can we infer concept similarity by measuring conditional complexity on the Web?

The "Google" distance



we can verify some simple properties.

The distance of a string with itself is close to 0, as expected.

Now, if I change the order of the letters, the distance increases, but not that much.

Same thing if I repeat the string multiple times.

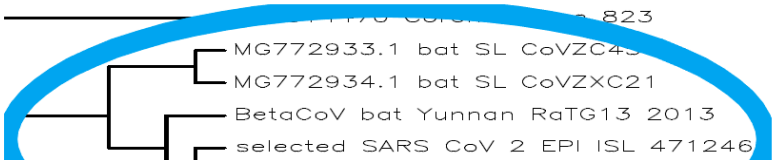


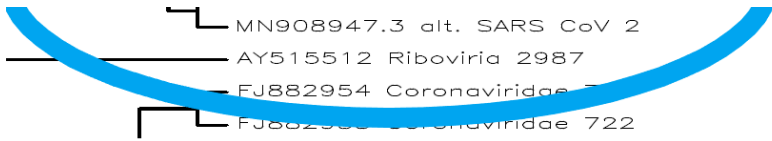
From NID to NCD. The Normalized Information Distance (NID) remains an abstract notion, based on Kolmogorov complexity. If we substitute a mere compressor such as "zip" for Kolmogorov Complexity, NID becomes NCD: the **Normalized Compression Distance** (here Z designates the length of the compressor output):

$$NCD(x, y) = \frac{Z(x, y) - \min[Z(x), Z(y)]}{\max[Z(x), Z(y)]}$$

The advantage of NCD is that it is computable and can be used in practice.

NCD has been used to compute distances between languages or distances between species based on DNA strings (Li, Chen & Li, 2004), and more recently to determine the phylogeny of the SARS-COV2 virus (Vitanyi & Cilibrasi, 2020). (see figure).





NCD

1/1 point (ungraded)

Consider this Python program and its execution.

```
import os, zlib

# Opening four sample texts
SampleFiles = ['sample1.txt', 'sample2.txt', 'sample3.txt', 'sample4.txt']
Texts = [open(F, 'rb').read() for F in SampleFiles]

# compressing texts
ZSizes = [len(zlib.compress(T)) for T in Texts]
print('Compressed sizes:\t', ZSizes)

# co-compressing texts
print('Z(sample1 + sample2):\t', len(zlib.compress(Texts[0] + Texts[1])))
print('Z(sample3 + sample4):\t', len(zlib.compress(Texts[2] + Texts[3])))
```

```
$ SampleCompress.py
Compressed sizes: [8247, 10793, 2151, 2829]
Z(sample1 + sample2): 18022
Z(sample3 + sample4): 4352
```

Which text pair should be regarded as more similar according to NCD?

- ☐ NCD(sample1, sample2) is smaller by 0.13
- ☒ NCD(sample3, sample4) is smaller by 0.13
- ☐ NCD(sample1, sample2) is smaller by 0.27
- ☐ NCD(sample3, sample4) is smaller by 0.27
- ☐ NCD(sample1, sample2) is smaller by 0.33
- ☐ NCD(sample3, sample4) is smaller by 0.33

✓

Answer
Correct:
 $NCD(sample1, sample2) = 0.91$
 $NCD(sample3, sample4) = 0.78$ so the answer is that $NCD(sample3, sample4)$ is smaller by 0.13.

Submit You have used 1 of 1 attempt

✓ Correct (1/1 point)

Question: Can we go one step further and use the Web to define a complexity-based distance

Web to define a complexity based distance between concepts?

Answer: Yes, and this is what Rudi Cilibrasi and Paul Vitányi did in 2007 by creating what they call the "Google distance".

From NCD to the "Google distance". We may use Web data to estimate K . Let's start from the expression of NID:

$$NID(x,y) = \frac{K(x,y)-\min[K(x),K(y)]}{\max[K(x),K(y)]}$$

As we did to get an estimate of conditional complexity, we may replace $K(x)$ by $\log_2(1/f(x))$, where $f(x)$ is the observed frequency of x on the Web, or rather by $\log_2(N/g(x))$, where N is the corresponding total number of indexed pages and $g(x)$ denotes the number of pages containing x .

$$NGD(x,y) = \frac{\max[\log(g(x)),\log(g(y))]-\log(g(x,y))}{\log(N)-\min[\log(g(x)),\log(g(y))]}$$

NGD stands for **Normalized Google Distance**, after the name of the search engine used by Cilibrasi and Vitányi. $g(x,y)$

denotes the number of pages containing both x and y , as

reported by the Web search engine (for Google, it seems that $N = 2 \times 10^{12}$ is a reasonable estimate, in relation to the declared number of hits).

Intuitively, NID is supposed to extract by whatever ideal means what is common to two objects. By estimating K using a Web search engine, as in NGD , the two objects are compared not through their common structural features, but through the human-created contexts in which they co-occur.

NID-NCD-NGD

0/1 point (ungraded)

Select all of the statements that are true.

☒ Conditional complexity is replaced by $K(x,y) - K(y)$ in NID.



☐ Normalization consists of dividing by standard deviation.

< Previous

Next >



edX

Legal

[Terms of Service & Honor Code](#)

[Privacy Policy](#)

[Accessibility Policy](#)

[Trademark Policy](#)

[Sitemap](#)

[Cookie Policy](#)

[Your Privacy Choices](#)

Connect

[Idea Hub](#)

[Contact Us](#)

[Help Center](#)

[Security](#)

[Media Kit](#)

